

The power W needed to move the propeller of a small airplane depends on the airplane velocity V , propeller diameter D , angular velocity Ω , and also on the air properties, including its density ρ , viscosity μ , and sound velocity c ; the latter, characterizing compressibility effects, has dimensions $[c] = LT^{-1}$. Assuming that the effect of gravity is negligible:

1. Use dimensional analysis to express the propeller power W as a function of the minimum number of dimensionless parameters.

A new airplane with propeller of diameter $D = 1.5 \text{ m}$ is to be designed to fly at an altitude $z = 5000 \text{ m}$ with cruise velocity $V = 300 \text{ Km/h}$ and propeller angular velocity $\Omega = 2500 \text{ rpm}$. To determine the needed power, an experimental model is to be tested in a wind tunnel at sea level ($z = 0 \text{ m}$).

2. Determine the model diameter D_e , the propeller angular velocity Ω_e , and the wind-tunnel velocity V_e needed to guarantee that the experiment provides a physically similar representation of the airplane flight conditions.
3. For the values of D_e , Ω_e , and V_e determined above, the power measured in the experiment (obtained as the product of Ω_e and the torque acting on the propeller shaft) is $W_e = 6 \times 10^7 \text{ Watt}$. Obtain the power W corresponding to the real airplane.

Note: Physical properties for air according to the standard atmosphere:

z m	T K	ρ kg/m ³	μ kg/(m s)	c m/s
0	288.15	1.225	1.775×10^{-5}	340.3
5000	255.65	0.736	1.630×10^{-5}	320.5

$$\begin{aligned}
 1) \quad W &= W(\rho, \mu, c, V, D, \Omega) \Rightarrow \frac{W}{\rho V^3 D^2} = f\left(\frac{\rho V D}{\mu}, \frac{V}{c}, \frac{\Omega D}{V}\right) \\
 2) \quad \frac{\rho V D}{\mu} &= \frac{\rho_e V_e D_e}{\mu_e} \Rightarrow \boxed{D_e = \frac{\mu_e}{\mu} \frac{\rho}{\rho_e} \frac{V}{V_e} D = \frac{1.630}{1.775} \frac{1.225}{0.736} \frac{300}{319} D = 0.924 \text{ m}} \\
 \frac{V}{c} &= \frac{V_e}{c_e} \Rightarrow \boxed{V_e = \frac{c_e}{c} V = 319 \text{ Km/h}} \\
 \frac{\Omega D}{V} &= \frac{\Omega_e D_e}{V_e} \Rightarrow \boxed{\Omega_e = \frac{D}{D_e} \frac{V_e}{V} \Omega = 4308 \text{ rpm}} \\
 3) \quad \frac{W}{\rho V^3 D^2} &= \frac{W_e}{\rho_e V_e^3 D_e^2} \Rightarrow \boxed{W = \frac{\rho}{\rho_e} \left(\frac{V}{V_e}\right)^3 \left(\frac{D}{D_e}\right)^2 W_e = 7.9 \cdot 10^7 \text{ Watt}}
 \end{aligned}$$